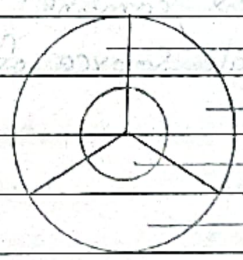
Our Environment

Environment
↓
Our Surrounding
- Artificial
- Natural
  - Abiotic → Non-living
  - Biotic → living

Ecosystem
↓
Intersection in our environment
- Abiotic & Abiotic → Temperature & Humidity
- Biotic & Biotic → Plants & Animals
- Biotic & Abiotic → Plants & Soil

Ecology – study of environment

Bio Sphere
↓ ↓
life zone which is circular



Biotic components of ecosystem:

Plant, Animals, Micro-organism, Bacteria, fungi, protozoa, Pathogens, Viruse, Producers - Prepare food, consumers - consume the food produced by producers.

Biotic forms of Ecosystem:-

1 Heterotrophs:

* Herbivore → Deer (prey) ↺ Hunt
* Carnivore → Tiger (predator)
* Omnivore
* Scavengers → Hyena, vulture
* Saprophyte → Ant -
  Physical breakdown of large particles into smaller forms.
* Decomposers → Bacteria -
  Convert complex organic into simple inorganic forms

Consumer = Consume food in form of energy.

Grass → grass hopper → frog → snake → peacock

↓ Producer (Produces energy for ecosystem)

↓ 1° cons

↓ 2° cons

↓ 3° cons

Food chain & Food web:-

Process of eating and being eaten in a food chain. Multiple food chains combine together forms food web. Sun is the ultimate source of food.

2 Autotrophs:

Sun → 1% → Autotrophs / → Uses for photosynthesis
(100% Energy) ↓ Green plants to prepare food
→ 99% waste

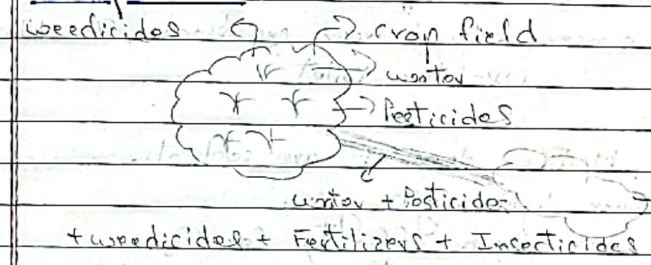
PAR → Photosynthetic Active Radiation.

Biomagnification - (Bio = living things, magnifi = multiply)

Green grass → Deer → Tiger
10 ppm 12 ppm 13 ppm

- The amount of toxicity increases within a food chain through various trophic levels.
- Toxicity is measured by ppm = Parts Per Million.
- There is no fixed amount of increase in toxicity.

Eutrophication:



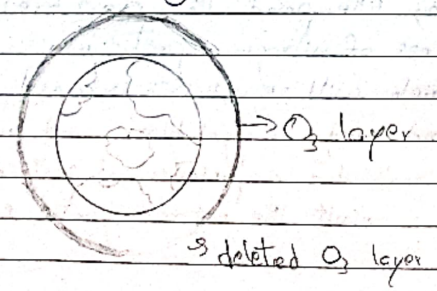
The extra water which is drained from nearby crop field will be passed into an aquatic water body like pond. This crop field water will contain excess of nitrogen, pesticides, insecticides, weedicides which will result into increase in nitrogen level of the pond leading to algal bloom and killing of other aquatic animals due to presence of pesticides, insecticides, etc. As a result the entire aquatic ecosystem will die.

Ozone layer and how it is depleted:-

ATMOSPHERE	Exosphere
	Ionosphere
	Mesosphere
	Stratosphere
	Troposphere
	Earth's Surface

Natural O_3 layer protects us from UV_A and UV_B radiation of Sun. UV_A and UV_B radiation causes skin cancer, vein disorder, eye disorder, Snow blindness.
UV-Ultra Voilet

- CFCs, HCFCs, Aerosoles are coolants.
- They are light wieght gases.
- They have high affinity for O_3.
- These gases reacts with O_3 causing depletion in O_3 layer and ultimately O_3 hole is created at southern polar region.



Artificial O_3 layer:

$C + O_2 \xrightarrow[\text{Complete combustion}]{} CO_2$

$C + \frac{1}{2}O_2 \xrightarrow[\text{Incomplete combustion}]{} CO + \frac{1}{2}O_2$

$\frac{1}{2}O_2 + O_2 \rightarrow O_3$ (prepared in troposphere)

O_3 in troposphere do not prevent UV_A and UV_B radiation.

- This leads air pollution and breathing disorder.

Managing the waste we produce

Living breakdown Biodegradable	Non-Biodegradable

Causes of waste production:

- Over population
- Irregular Sanitation
- More Industrilisation
- Improper waste Management
- Over exploitation
- Change in life style